Ontology Engineering

Session 3

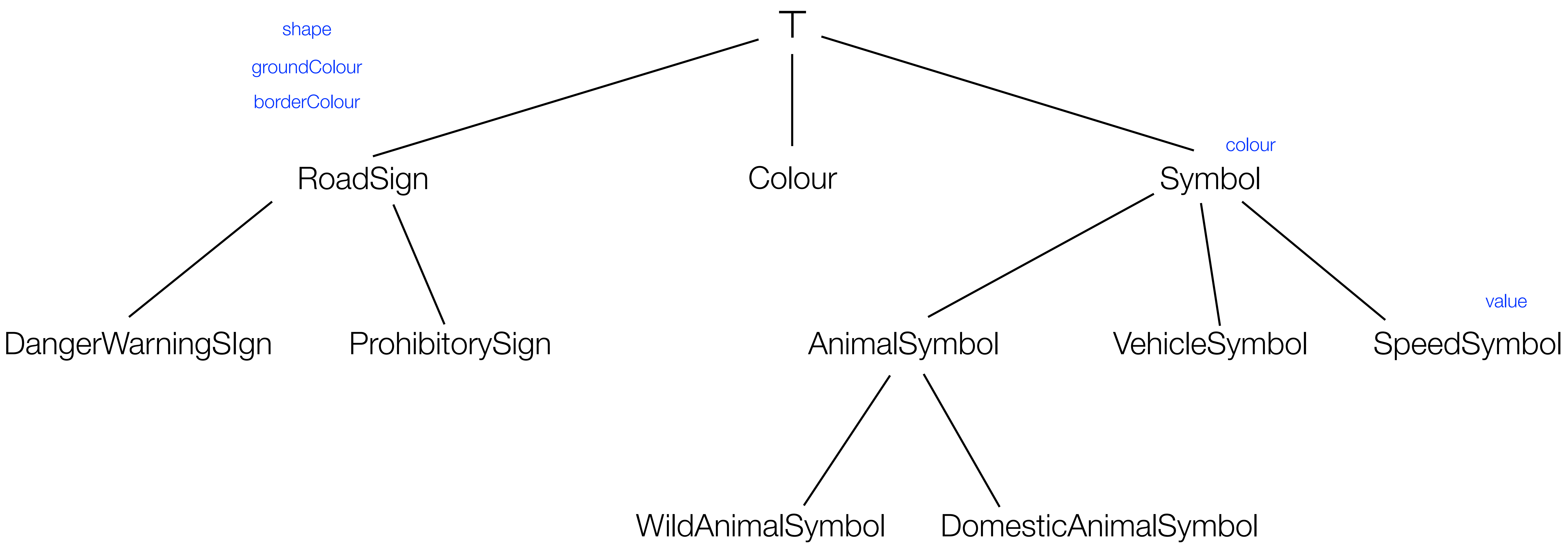
Assumptions

- Febraury 18
  - Protégé installed, and basic knowledge of its functionalities
- February 25
  - formal taxonomy of road signs as an IS-A class hierarchy, written in OWL2
  - conformant to Part I, Annex 1, of Vienna Convention (document "Conv_road_signs_2006v_EN.pdf")
- March 4
  - extended taxonomy, aligned with the category labels of the RDF knowledge graph of the Mapillary Traffic Sign Dataset (file "MTSD.ttl")

A Simple Ontology Development Methodology

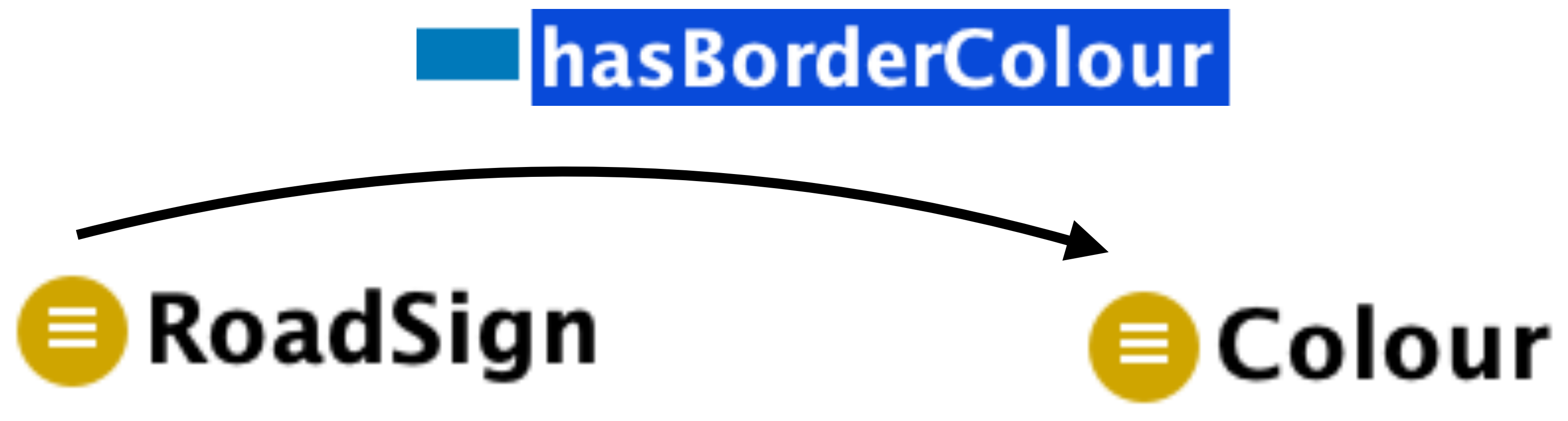
1. Determine the domain and scope of the ontology
2. Consider reusing existing ontologies
3. Enumerate terms in the ontology
4. Define the classes and the class hierarchy
5. Define the properties of classes
6. Define the characteristics of properties
7. Create instances

Step 5: Define the Properties of Classes

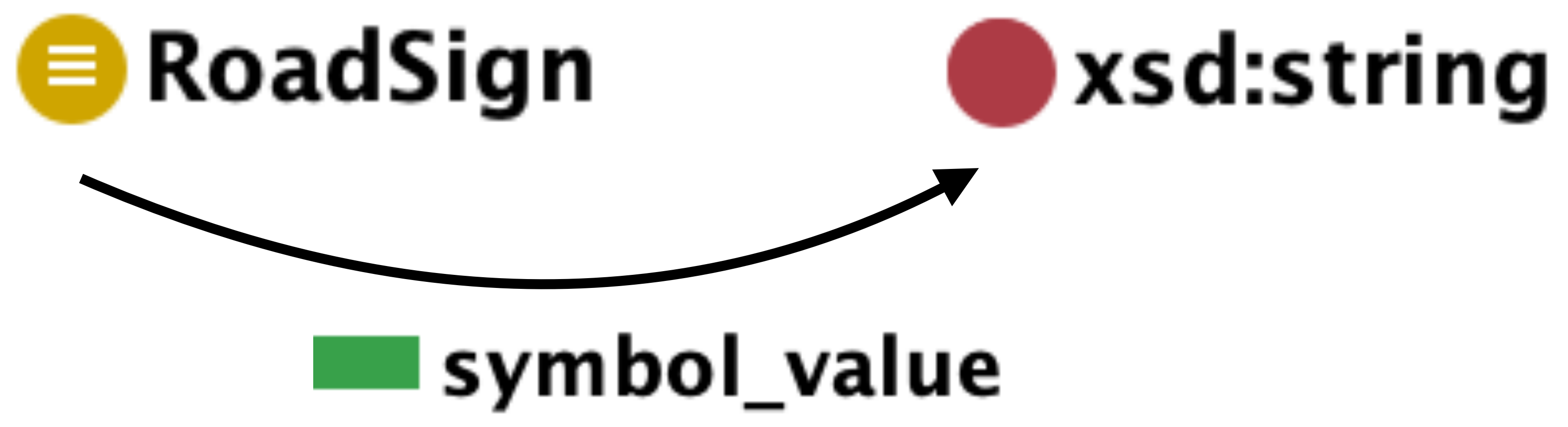


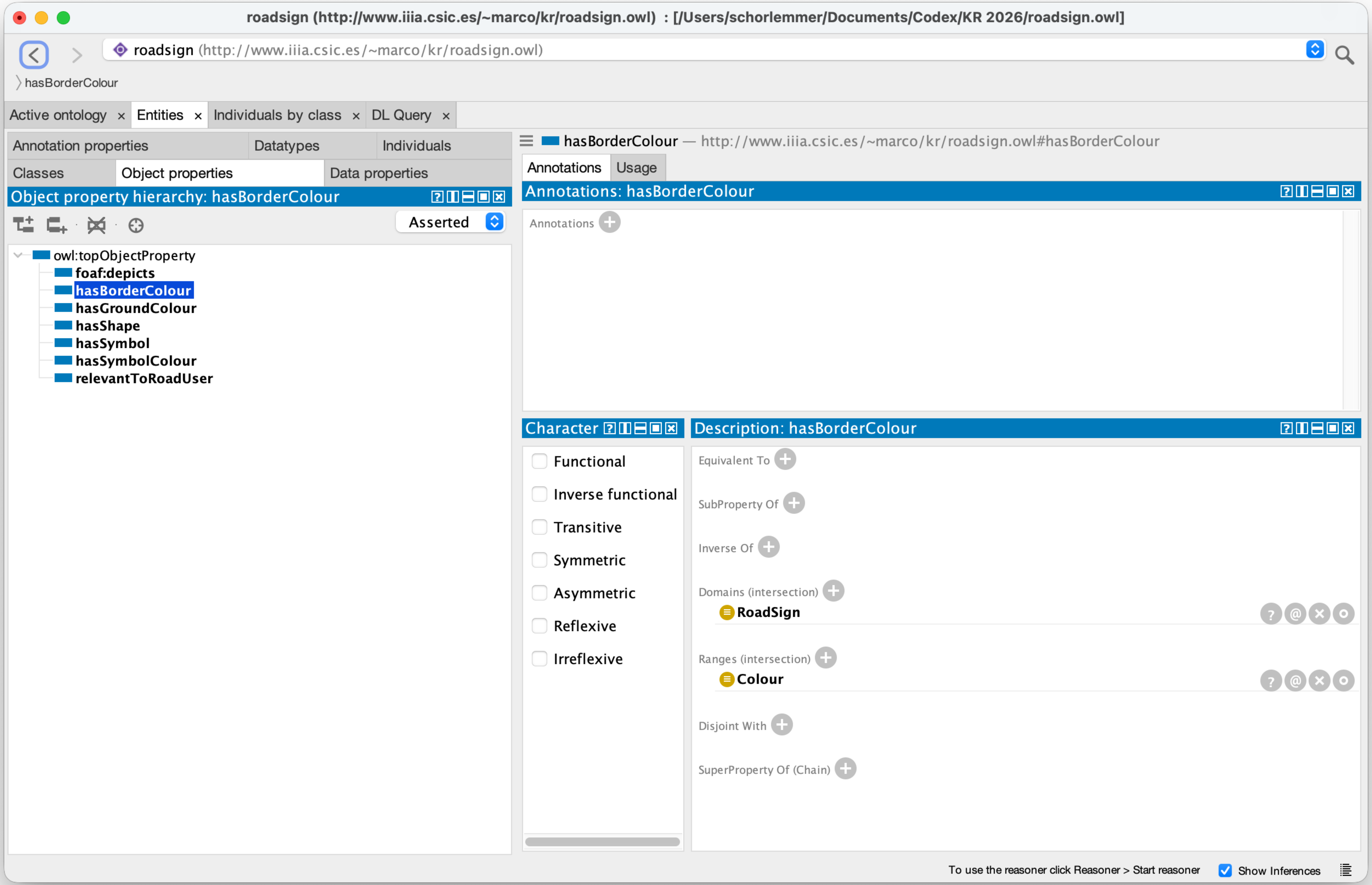
Object and Data Properties

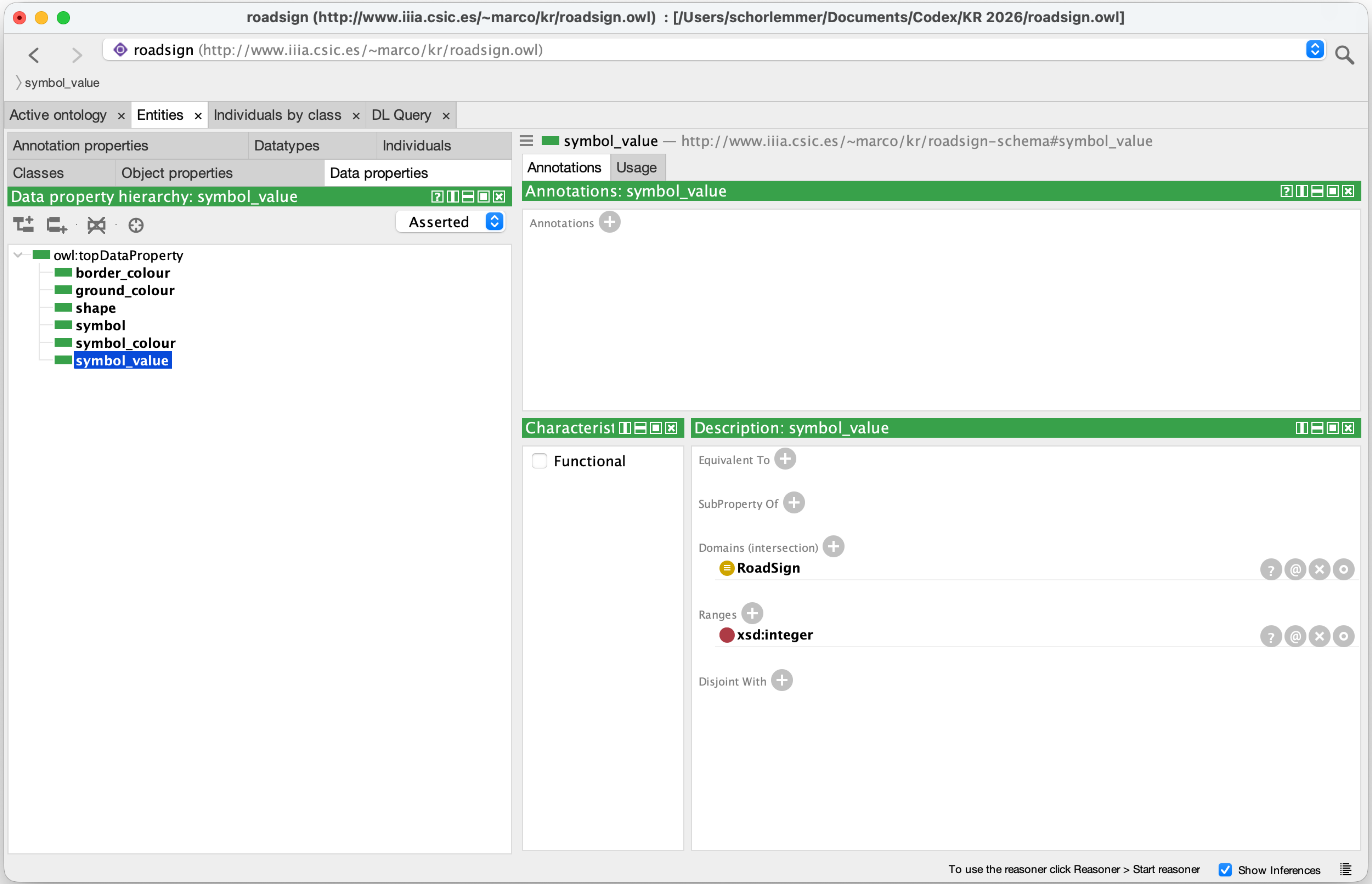
- Object properties
  - relations between two classes



- Data properties
  - relations between a class and a datatype




roadsign (http://www.iiia.csic.es/~marco/kr/roadsign.owl) : [/Users/schorlemmer/Documents/Codex/KR 2026/roadsign.owl]
roadsign (http://www.iiia.csic.es/~marco/kr/roadsign.owl)
hasBorderColour
Active ontology
Entities
Individuals by class
DL Query
Annotation properties
Datatypes
Individuals
Classes
Object properties
Data properties
Object property hierarchy: hasBorderColour
Asserted
owl:topObjectProperty
foaf:depicts
hasBorderColour
hasGroundColour
hasShape
hasSymbol
hasSymbolColour
relevantToRoadUser
hasBorderColour — http://www.iiia.csic.es/~marco/kr/roadsign.owl#hasBorderColour
Annotations
Usage
Annotations: hasBorderColour
Annotations
Character
Functional
Inverse functional
Transitive
Symmetric
Asymmetric
Reflexive
Irreflexive
Description: hasBorderColour
Equivalent To
SubProperty Of
Inverse Of
Domains (intersection)
RoadSign
Ranges (intersection)
Colour
Disjoint With
SuperProperty Of (Chain)
To use the reasoner click Reasoner > Start reasoner
Show Inferences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symbol_value
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Data properties
Data property hierarchy: symbol_value
Asserted
owl:topDataProperty
border_colour
ground_colour
shape
symbol
symbol_colour
symbol_value
symbol_value — http://www.iiia.csic.es/~marco/kr/roadsign-schema#symbol_value
Annotations
Usage
Annotations: symbol_value
Annotations
Characterist
Description: symbol_value
Functional
Equivalent To
SubProperty Of
Domains (intersection)
RoadSign
Ranges
xsd:integer
Disjoint With
To use the reasoner click Reasoner > Start reasoner
Show Inferences

Assignment

- Add to your ontology (which currently only holds a class hierarchy):
    - **object properties**, with their **domain** and **range**, covering the attributes of the road-sign classes you have defined so far, as described in Part I, Annex 1 of the document “Conv_road_signs_2006v_EN.pdf"
    - **data properties**, with their **domain** and **range**,
        - covering the attributes of the road-sign classes you have defined so far, as described in Part I, Annex 1 of the document “Conv_road_signs_2006v_EN.pdf" (e.g., to know the value of speed limits, maximum heights or weights of vehicles, textual information given on information or direction signs, etc.)
        - covering additional attributes of the MTSD dataset (.e.g, relevant metadata information of the images in which the road signs are depicted)
- Do this using the appropriate prompts for LLM-based ontology development
  (Recommendation: do it in two steps, first object, then data properties)
    - you may accompany the prompt with competency questions about road-sign properties and MTDS metadata

Example of Prompt Template

Role: You are an expert Ontology Engineer specialising in [generic domain of the ontology]

Objective: [describe the task to do]

Scope & Domain:

- Domain: [define the concrete scope of the domain to be modelled]
- Key Entities: [provide key entities that need to be included in the ontology]
- Boundaries: [define what NOT to include to prevent scope creep]

Competency Questions: The ontology must be able to answer the following questions:

- [Competency Question 1]
- [Competency Question 2]
- …

Structural Requirements: [describe particular implementation requirements for this task]

Output Format: [specify the concrete output format (XML/RDF, Turtle, …)

Remember to Keep a Record of…

- For each engineering step:
    - LLMs used
    - prompt (or series of prompts)
    - revised output file
- This information has to be provided in the final submission